

MOSTAFA ZAMAN

*Ph.D. Candidate in Electrical and Computer Engineering
Virginia Commonwealth University*

Phone: 804-938-5784 | Email: zamanm@vcu.edu | Website: occtestbed.com

LinkedIn: [mostafa-zaman](#) | GitHub: [Naabiil-09](#)

EDUCATION

Virginia Commonwealth University

Doctor of Philosophy (Ph.D.) in Electrical and Computer Engineering

December, 2022 – Ongoing

Richmond, VA, USA

Virginia Commonwealth University

Master of Science (MSc) in Electrical and Computer Engineering

August, 2018 – December, 2022

Richmond, VA, USA

Bangladesh University of Engineering and Technology

Bachelor of Science (BSc) in Electrical and Electronic Engineering

May, 2010 – September, 2015

Dhaka, Bangladesh

RESEARCH EXPERIENCE

Development of the OpenCyberCity Smart-City Testbed

Graduate Research Assistant, Virginia Commonwealth University

August, 2018 – Present

Richmond, VA, USA

- Develop and maintain the OpenCyberCity smart-city cyber-physical testbed spanning smart buildings, transportation, manufacturing, and healthcare domains.
- Design control, optimization and data-driven analytics (machine learning) pipelines to support real-time monitoring and decision support in smart-city applications.
- Build bi-level optimization and reinforcement-learning components for interconnected complex smart-city domains, emphasizing constraint-aware and uncertainty-informed decision making.
- Integrate sensing and hardware components for the testbed (sensor integration and building subsystem interfaces).
- Contribute to dissemination and proposal development through peer-reviewed publications, posters, and grant-related activities.

RESEARCH INTERESTS

- Smart Cities and Urban Systems.
- Cyber-Physical Systems Design and Integration.
- Machine Learning and Artificial Intelligence for Control & Optimization.
- Deep Learning and Reinforcement Learning.
- Adaptive Control Systems under Uncertainty
- Internet of Things (IoT) and Sensor Networks

PEER-REVIEWED PUBLICATIONS

Journal Paper 1

June, 2024

MDPI

- * [J1] **M. Zaman**, N. Puryear, S. Abdelwahed and N. Zohrabi, 2024. A review of IoT-based smart city development and management. *Smart Cities*, 7(3), pp.1462-1501.

Conference Paper 1

September, 2017

Dhaka, Bangladesh

- * [C1] M. R. H. Rashed, **M. Zaman**, Mobassear-Ul-Islam and M. F. Raihan, "An analysis on the required reinforcement for embedding a nuclear power plant in a generic power system", 2017 4th International Conference on Advances in Electrical Engineering (ICAEE), 2017, pp. 597-600, doi: 10.1109/ICAEE.2017.8255427.

Conference Paper 2

June, 2019

Richmond, VA, USA

- * [C2] A. Morrissett, R. Eini, **M. Zaman**, N. Zohrabi and S. Abdelwahed, "A Physical Testbed for Intelligent Transportation Systems", 2019 12th International Conference on Human System Interaction (HSI), 2019, pp. 161-167, doi: 10.1109/HSI47298.2019.8942615.

Conference Paper 3

June, 2020

Chicago, Illinois, USA

- * [C3] **M. Zaman**, N. Zohrabi and S. Abdelwahed, "A CNN-based Path Trajectory Prediction Approach with Safety Constraints", 2020 IEEE Transportation Electrification Conference & Expo (ITEC), 2020, pp. 267-272, doi: 10.1109/ITEC48692.2020.9161731.

Conference Paper 4

September, 2021

Manchester, United Kingdom

- * [C4] N. Zohrabi, P. J. Martin, M. Kuzlu, L. Linkous, R. Eini, A. Morrissett, **M. Zaman** et al., "OpenCity: An Open Architecture Testbed for Smart Cities", 2021 IEEE International Smart Cities Conference (ISC2), 2021, pp. 1-7, doi: 10.1109/ISC253183.2021.9562813.

Conference Paper 5

September, 2021

Manchester, United Kingdom

- * [C5] **M. Zaman**, M. Al Islam, A. Tantawy, C. J. Fung and S. Abdelwahed, "Adaptive Control for Smart Water Distribution Systems", 2021 IEEE International Smart Cities Conference (ISC2), 2021, pp. 1-6, doi: 10.1109/ISC253183.2021.9562812.

Conference Paper 6

November, 2021

Long Beach, CA, USA

- * [C6] **M. Zaman**, S. Saha, R. Eini & S. Abdelwahed, "A Deep Learning Model for Forecasting Photovoltaic Energy with Uncertainties", 2021 IEEE Green Energy and Smart Systems Conference (IGESSC), 2021, pp. 1-6, doi: 10.1109/IGESSC53124.2021.9618681.

Conference Paper 7

December, 2021

Haikou, Hainan, China

- * [C7] N. Puryear, **M. Zaman**, R. Eini & S. Abdelwahed, "Design and Implementation of a Distributed Control Platform for a Smart Building Testbed", IEEE 19th International Conference on Smart City, 2021, pp. 1443-1450, doi: 10.1109/HPCC-DSS-SmartCity-DependSys53884.2021.00216.

Conference Paper 8

June, 2022

Anaheim, CA, USA

- * [C8] **M. Zaman**, S. Saha, N. Puryear, N. Zohrabi & S. Abdelwahed, "Incorporation of Physiological Features in Drowsiness Detection Using Deep Neural Network Approach", 2022 IEEE Transportation Electrification Conference & Expo (ITEC), Anaheim, CA, USA, 2022, pp. 219-224, doi: 10.1109/ITEC53557.2022.9813946.

Conference Paper 9

August, 2022

Vancouver, BC, Canada

- * [C9] **M. Zaman**, S. Saha, N. Zohrabi and S. Abdelwahed, "Uncertainty Estimation in Power Consumption of a Smart Home Using Bayesian LSTM Networks", 2022 IEEE International Symposium on Advanced Control of Industrial Processes (AdCONIP), Vancouver, BC, Canada, 2022, pp. 120-125, doi: 10.1109/AdCONIP55568.2022.9894187.

Conference Paper 10

September, 2022

Pafos, Cyprus

- * [C10] **M. Zaman**, R. Eini, N. Zohrabi and S. Abdelwahed, "A Decision Support System for Cyber Physical Systems under Disruptive Events: Smart Building Application", 2022 IEEE International Smart Cities Conference (ISC2), Pafos, Cyprus, 2022, pp. 1-7, doi: 10.1109/ISC255366.2022.9922493.

Conference Paper 11

October, 2022

Boston, USA

- * [C11] **M. Zaman**, M. A. Islam, A. Tantawy and S. Abdelwahed, "An Uncertainty Based Predictive Analysis of Smart Water Distribution System Using Bayesian LSTM Approach", 2022 6th International Conference on Universal Village (UV), Boston, MA, USA, 2022, pp. 1-7, doi: 10.1109/UV56588.2022.10185529.

Conference Paper 12

June, 2023

Detroit, MI, USA

- * [C12] **M. Zaman**, S. Saha and S. Abdelwahed, "Assessing the Suitability of Different Machine Learning Approaches for Smart Traffic Mobility", 2023 IEEE Transportation Electrification Conference & Expo (ITEC), Detroit, MI, USA, 2023, pp. 1-6, doi: 10.1109/ITEC55900.2023.10186901.

Conference Paper 13

June, 2023

Detroit, MI, USA

- * [C13] **M. Zaman**, S. Saha, N. Zohrabi and S. Abdelwahed, "Deep Learning Approaches for Vehicle and Pedestrian Detection in Adverse Weather", 2023 IEEE Transportation Electrification Conference & Expo (ITEC), Detroit, MI, USA, 2023, pp. 1-6, doi: 10.1109/ITEC55900.2023.10187020.

Conference Paper 14

December, 2023

Florida, USA

- * [C14] **M. Zaman**, A. Tantawy and S. Abdelwahed, "Optimizing Smart City Water Distribution Systems Using Deep Reinforcement Learning", 2023 IEEE 20th International Conference on Smart Communities: Improving Quality of Life using AI, Robotics and IoT (HONET), Boca Raton, FL, USA, 2023, pp. 228-233, doi: 10.1109/HONET59747.2023.10374894.

Conference Paper 15

December, 2023

Florida, USA

- * [C15] **M. Zaman**, N. Puryear, A. Malik and S. Abdelwahed, "Emulation of Smart Grid Technologies and Topologies in a Small Scale Smart City Testbed", 2023 IEEE 20th International Conference on Smart Communities: Improving Quality of Life using AI, Robotics and IoT (HONET), Boca Raton, FL, USA, 2023, pp. 234-239.

Conference Paper 16

February, 2024

Washington DC, USA

- * [C16] **M. Zaman**, S. Saha, N. Zohrabi and S. Abdelwahed, "Demand-Response Prediction in Smart Grids Using Machine Learning Techniques", 2024 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT), Washington, DC, USA, 2024, pp. 1-5, doi: 10.1109/ISGT59692.2024.10454224.

Conference Paper 17

July, 2024

Osaka, Japan

- * [C17] **M. Zaman**, M. Al Islam, N. Zohrabi and S. Abdelwahed, "A Machine Learning-Based Temperature Control and Security Protection for Smart Buildings", 2024 IEEE International Conference on Smart Computing (SMARTCOMP), Osaka, Japan, 2024, pp. 290-295, doi: 10.1109/SMARTCOMP61445.2024.00070.

Conference Paper 18

August, 2025

Calgary, Canada

- * [C18] **M. Zaman**, A. Malik, D. Gubay, A. DeWitt, S. Varzaneh, N. Zohrabi, S. Abdelwahed, "Integrative ITS Platform for Enhanced Urban Mobility in Smart Cities", IEEE International Conference on Ubiquitous Intelligence and Computing, Calgary, Alberta, Canada, 2025

Conference Paper 19

April, 2026

California, USA

- * [C19] **M. Zaman**, S. Iannucci, N. Zohrabi and S. Abdelwahed, "A Multi-Level Reinforcement Learning Framework for Smart Building Management", Accepted at 13th IEEE Conference on Technologies for Sustainability, 2026 (SusTech 2026)

Conference Paper 20

April, 2026

California, USA

- * [C20] S. Varzaneh, **M. Zaman**, N. Zohrabi and S. Abdelwahed, "A Review of AI-based Optimization and Monitoring Using Wearable Data in Smart Buildings", Accepted at 13th IEEE Conference on Technologies for Sustainability, 2026 (SusTech 2026)

ABSTRACTS & POSTERS

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| An Adaptive Smart Water Distribution Control System for the OpenCity Testbed
<i>Presented to CCI Symposium</i> | April, 2022
<i>Richmond, VA, USA</i> |
| Development of the OpenCyberCity Testbed: Smart City Research ... Opportunities
<i>Accepted to ICCPS-WIP</i> | May, 2023
<i>San Antonio, Texas USA</i> |
| Using Innovations in Data Analytics and Smart Technologies ... Opioid Overdose Crisis
<i>Accepted to Smartcomp2023</i> | June, 2023
<i>Nashville, Tennessee</i> |
| OpenCyberCity Testbed's Recent Progress in Smart City Management
<i>Best Poster Award</i> | June, 2024
<i>Osaka, Japan</i> |
| OpenCyberCity: Smart City Research and Solutions for Urban Development
<i>Accepted to CCI Symposium</i> | April, 2025
<i>Richmond, VA, USA</i> |
| Open Cyber City: A Scalable Cyber Physical Testbed ... Robotics Research
<i>Accepted to VRI Symposium</i> | November, 2025
<i>Charlottesville, VA, USA</i> |
| A Modular CPS City Testbed for End-to-End Sensing, Analytics, and Edge Intelligence
<i>Accepted to CCI Symposium</i> | April, 2026
<i>Richmond, VA, USA</i> |

TECHNICAL SKILLS

- **Programming Languages:** Python, MATLAB, C, LaTeX, JavaScript
- **Machine Learning/AI Frameworks:** TensorFlow, PyTorch, Scikit-learn, Keras, Deep Learning Toolbox
- **Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn
- **Development Tools & Environments:** Jupyter Notebook, Visual Studio, Git/GitHub, HPC Cluster
- **Simulation & Design Tools:** Simulink, PSpice, Vivado, Proteus, OpenMarkov, Circuitverse, Visio, PSAF
- **IoT & Embedded Systems:** Sensor Integration, FPGA Development, Microcontroller Programming
- **Other:** LaTeX, Mendeley, Draw.io, Microsoft Office Suite

TEACHING EXPERIENCE

Teaching Assistant <i>Virginia Commonwealth University</i>	August, 2018 – Present <i>Richmond, VA, USA</i>
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Courses Taught:

- **EGRE 335: Signals and Systems** (Spring 2026) - Graded homework and exams on linear continuous/discrete-time systems, convolution, Fourier series/transform, Laplace and Z transforms, and continuous-to-discrete conversion.
- **EGRE 364: Microcomputer Systems** (Fall 2022) - Assisted students with Vivado software for real-time applications; helped students understand simulation models and hardware for computer system functions; assessed lab reports and proctored exams.
- **EGRE 254: Digital Logic Design** (5 semesters: Fall 2019, Spring 2020, Fall 2020, Spring 2021, Fall 2022) - Guided students in designing and constructing digital logic circuits using discrete logic and field-programmable gate arrays (FPGAs); graded lab reports and proctored exams.
- **EGRE 245: Engineering Programming** (5 semesters: Fall 2018, Spring 2019, Fall 2019, Spring 2020, Spring 2022) - Tutored students in C-based structured programming fundamentals, problem-solving, algorithm design, and C syntax; assessed lab reports and proctored exams.
- **EGRE 553: Industrial Automation** (Fall 2020) - Assisted with programmable logic controllers (PLCs), supervisory control and data acquisition (SCADA), and human-machine interfaces (HMIs); graded lab reports and proctored exams.

- **EGRE 365: Digital Systems** (Spring 2019) - Supported VHDL modeling, simulation, synthesis, and FPGA design; assisted with microprocessor peripherals, interfacing, and embedded systems design; graded lab reports and proctored exams.
- **EGRE 246: Advanced Engineering Programming** (Fall 2018) – Tutored intermediate and advanced level C programming for engineering applications, including program design, problem-solving, and software testing; evaluated lab reports and proctored exams.

SUBMITTED & UNDER REVIEW

- [S1] M. Zaman and S. Abdelwahed Accepted, 2026
- * **“Graph-Based Bi-Level Reinforcement Learning for Coordinated EV Fast Charging Under Distribution Constraints,”** *Accepted to IEEE Transportation Electrification Conference & Expo (ITEC) 2026.*
- [S2] G. Addala, M. Zaman, N. Zohrabi, and S. Abdelwahed Submitted, 2026
- * **“An Integrated Framework for Forecast-Driven Priority-Aware Energy Allocation in Microgrids,”** *Submitted to IEEE Conference on Control Technology and Applications (CCTA) 2026.*
- [S3] T. Tahi, S. Saha, M. Zaman, Y. Zhao, and S. Abdelwahed Accepted, 2026
- * **“Context-Aware Functional Security for V2X-Enabled Connected Vehicles,”** *Accepted to IEEE Transportation Electrification Conference & Expo (ITEC) 2026.*

GRANTS & FUNDING PARTICIPATION

- CCI CVN Smart City Testbed Phase 2** August, 2022 – Present
Graduate Research Assistant Richmond, VA, USA
- * Supported development and expansion of the OpenCyberCity testbed, including new services, transportation components, and data/analytics capabilities
- NSF Project: Using Innovations in Sensing, Data Analytics ... Opioid Crisis** December, 2022 – July, 2023
Graduate Research Assistant Richmond, VA, USA
- * Contributed to planning activities to address opioid overdose crisis involving sensing/data collection and predictive/explainable analytics to support community-informed decision making
- VCU and Meru Biotech Collaborative Research** January, 2022 – August, 2022
Graduate Research Assistant Richmond, VA, USA
- * Supported development and evaluation of a collaborative sensing system to quantify differences in light refraction between fluid samples for molecular-binding detection applications
 - * Conducted data analysis and experimental validation to compare sample responses against control measurements

PROFESSIONAL SERVICE

Reviewer Activities

Journals & Conferences

- * IEEE Transactions on Artificial Intelligence
- * IEEE Open Journal of Control Systems
- * Frontiers in Energy Research – Smart Grids
- * Journal of Ambient Intelligence and Humanized Computing
- * IEEE Conference on Control Technology and Applications (CCTA) 2023
- * IEEE Green Energy and Smart Systems Conference
- * IEEE International Smart Cities Conference (ISC2)
- * IEEE International Conference on Smart Communities: Improving Quality of Life Using ICT, IoT & AI
- * IEEE CICN (International Conference on Computational Intelligence and Communication Networks)
- * International Conference on High-Performance Computing & Simulation
- * International Conference on Communication Systems and Network Technologies (CSNT)
- * International Conference on Computer Science and Application Engineering
- * International Conference on Computer Technology and Information Science

Professional Memberships

IEEE and Affiliated Societies

2022 – Present

- * IEEE Member
- * IEEE Young Professionals
- * IEEE Control Systems Society
- * IEEE Power Electronics Society
- * IEEE Systems Council

HONORS & AWARDS

- Awarded "Best Graduate Research Student" recognition by VCU Engineering for outstanding research performance (2024–2025).
- Best Poster Award, IEEE Smartcomp 2024, Osaka, Japan (June 2024).
- Travel Award, IEEE Transportation Electrification Conference & Expo (ITEC + EATS) (2022).
- Travel Awards, ECE Department and VCU College of Engineering (2021–2022).
- Recipient of Bangladesh Sweden Trust Fund (2018–2019).
- Dean's List, 4th Year Undergraduate.

PROFESSIONAL EXPERIENCE

Energypac Engineering Limited

March, 2016 – June, 2018

Assistant Engineer (R&D), Power Transformer Division

Dhaka, Bangladesh

- * Designed 132/11 kV and 33/11 kV power transformers including general arrangements (GA), guaranteed technical particulars (GTP), and nameplates .
- * Managed core-coil assembly, conductor and core requisitions, OLTC specifications, and winding details .
- * Prepared costing sheets and technical documentation for transformer manufacturing projects .
- * Completed professional training on "Safety of OLTC and Protection of Transformer" by ABB (Sweden) in August 2017 .

REFERENCES

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| • Dr. Sherif Abdelwahed
Professor
Department of Electrical &
Computer Engineering
Virginia Commonwealth University
Richmond, Virginia, 23220, USA
sabelwahed@vcu.edu | • Dr. Robert H. Klenke
Chair, Professor
Department of Electrical &
Computer Engineering
Virginia Commonwealth University
Richmond, Virginia, 23220, USA
rhklenke@vcu.edu | • Dr. Nasibeh Zohrabi
Assistant Professor
Department of Engineering
Pennsylvania State University
Media, Pennsylvania, 19063 USA
nmz5171@psu.edu |
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